WEEK 11 · Discrete Regimes

HMMs and Market Regimes

Estimating hidden market regimes as probabilities

Markov chains · Transition and emission · Forward · Viterbi · Baum–Welch · Risk budget

Today's question — is the market calm today, or in a high-risk regime?
Can we express it as a probability rather than a certainty?

An HMM estimates hidden regimes as probabilities

Three messages that run through this lecture

1

The hidden state is one of a few categories

Kalman's level and beta are continuous, but an HMM state is a discrete category such as calm or high-volatility.

2

Regime probabilities: predict by transition, correct by observation

Not from transition probabilities alone — it also weighs how well today's observation fits each regime.

3

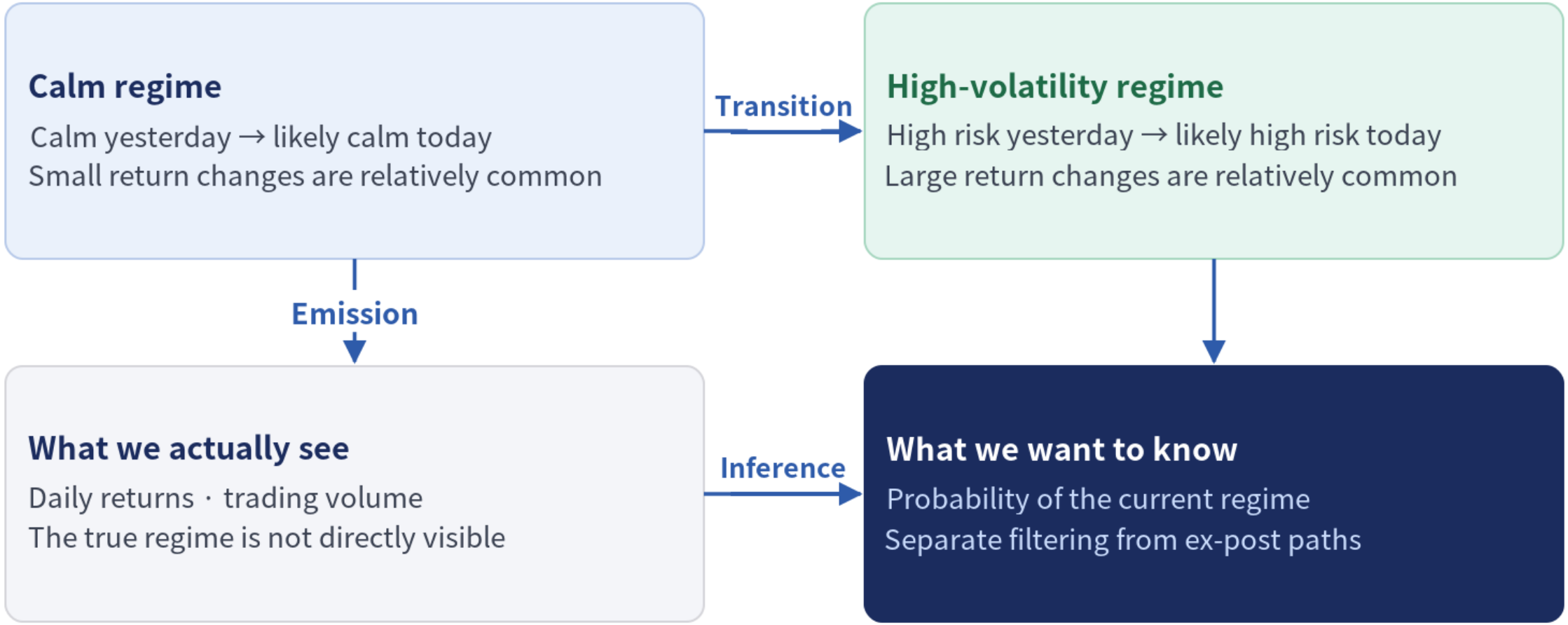
Each question has its own algorithm

Real-time probability: filtering; reinterpreting the past: smoothing; most likely path: Viterbi; parameters: Baum–Welch.

Never use full-sample smoothing or Viterbi paths as historical trading signals

We see returns; we want to know the regime

Regime names are economic interpretations; a high-volatility probability is not a probability of falling tomorrow



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PART 1 · PRINCIPLES

Markov Chains and Observation Distributions

Markov property · Transition matrix · Duration · Emission distribution

Markov property: the current state alone sets the next

A teaching example with states C (calm) and V (high volatility) — the names are just definitions

$$P(s_t \mid s_{t-1}, s_{t-2}, \ldots) = P(s_t \mid s_{t-1})$$

Meaning

- Given the current state, older states add no information about the next state

Do not misread

- It is a model assumption, not a claim that real markets remember only one day

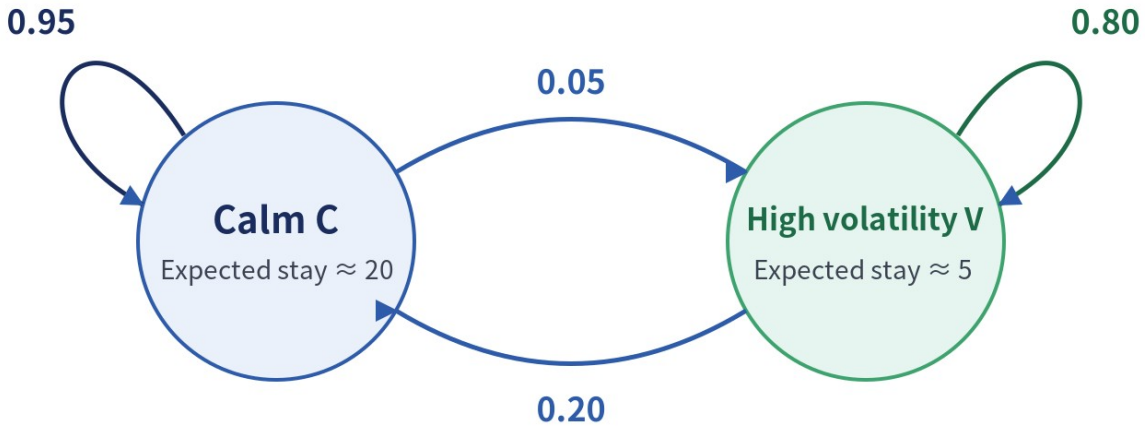
Transition matrix: probabilities from today's regime to tomorrow's

Rows are the current state, columns the next state — order C, V

$$\underbrace{A = \begin{bmatrix} 0.95 & 0.05 \\ 0.20 & 0.80 \end{bmatrix}}_{\text{transition matrix}}$$

How to read

- Calm → calm 95%
- High volatility → high volatility 80%
- Each row sums to 1



The higher the self-transition probability, the longer a regime lasts

Expected duration starting from state i in a standard time-homogeneous HMM

$$\underbrace{E[D_i]}_{\text{expected duration of state } i} = \frac{1}{1-a_{ii}}$$

In the example

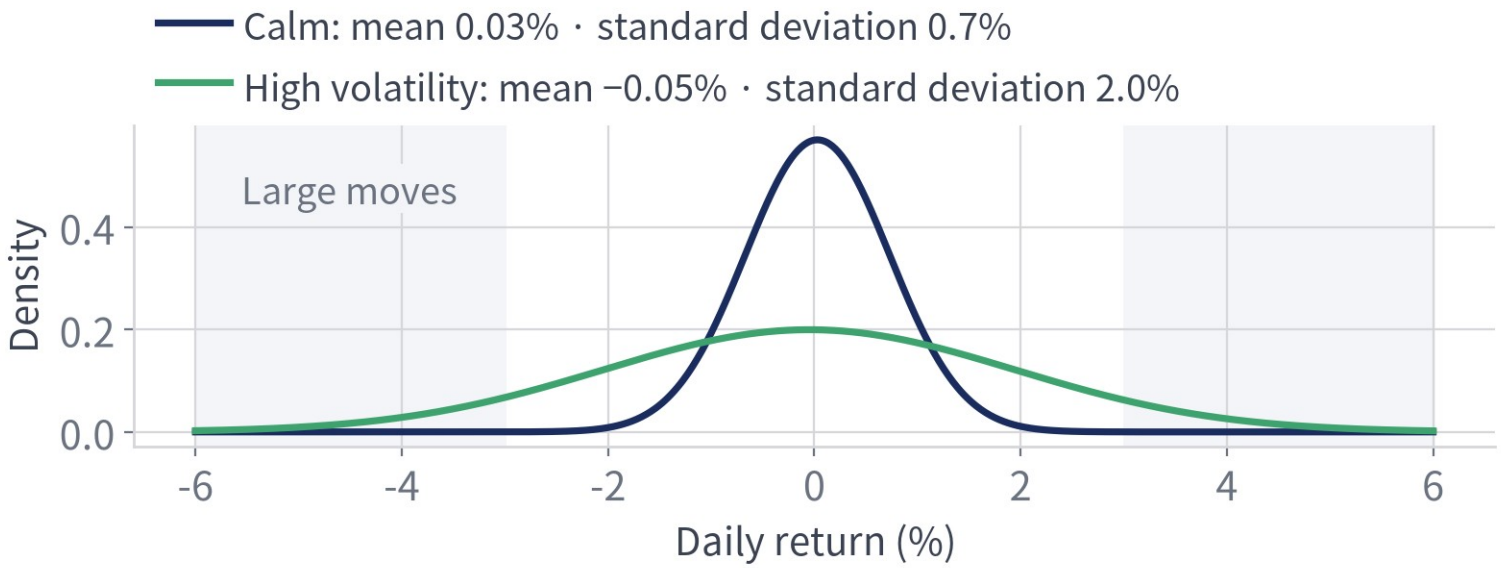
- Calm: 1 / (1 − 0.95) ≈ 20 observations
- High volatility: 1 / (1 − 0.80) ≈ 5 observations

Caution

- It is not a fixed end date
- With daily data, the unit is trading days

Each hidden regime generates returns from a different distribution

Given the regime, returns are drawn from that regime's normal distribution · hypothetical parameters



Interpretation

Prices can rise in a high-volatility regime too

Naming

‘Crisis’ is an ex-post reading · order state labels by mean and variance

Given the regime, returns follow a normal distribution

Emission distribution — state j has its own mean and variance

$$r_t \mid s_t = j \sim \mathcal{N}(\mu_j, \sigma_j^2)$$

Hypothetical parameters

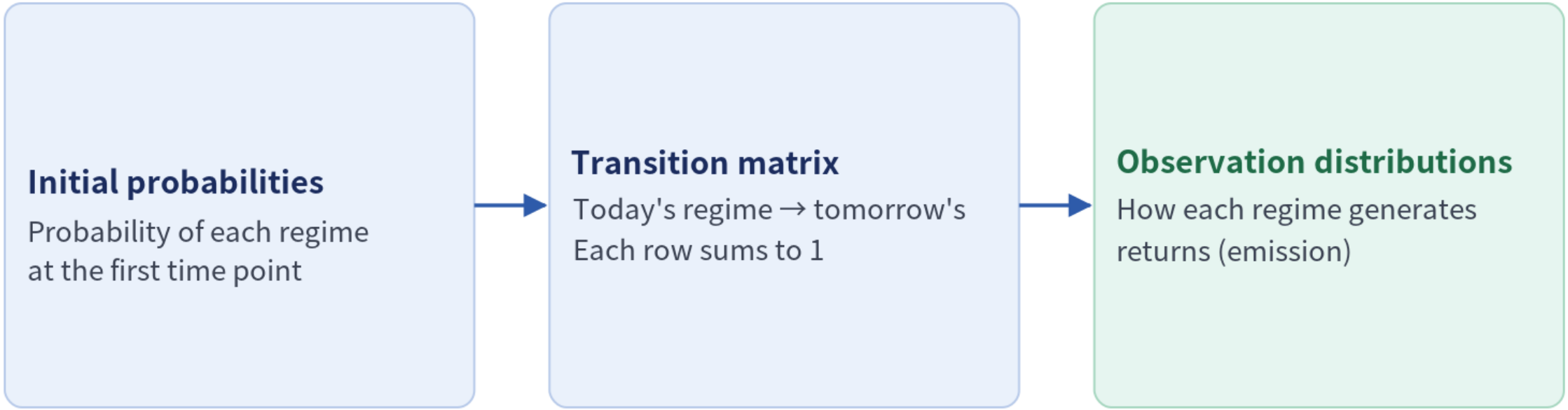
- Calm: daily mean 0.03% · standard deviation 0.7%
- High volatility: daily mean −0.05% · standard deviation 2.0%

Standard assumption and extensions

- Given the current state, the current observation is conditionally independent of other observations
- Regime-specific AR dynamics are a separate extension: Markov-switching AR

An HMM is fully specified by three components

Initial probabilities · Transition matrix · State-specific observation distributions



Label switching

State labels can swap at each re-estimation, so order them by mean and variance

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PART 2 · ALGORITHMS

Updating Regime Probabilities and Estimating Paths

Filtering · Forward · Viterbi · Baum–Welch

Predict by transition, correct by observation

Start from the state probabilities filtered up to yesterday · the emission density is the density of today's observation under state j

$$\underbrace{q_t(j) = \sum_i p_{t-1}(i) a_{ij}}_{\text{before observing: predict by transition}}$$

$$\underbrace{p_t(j) = \frac{b_j(r_t) q_t(j)}{\sum_k b_k(r_t) q_t(k)}}_{\text{after observing: correct by emission density}}$$

Two steps

- Before observing: predict today's regime probabilities with the transition matrix
- After observing: weight by each regime's emission density and normalize

Caution

- A density value for continuous returns is not a point probability and can exceed 1

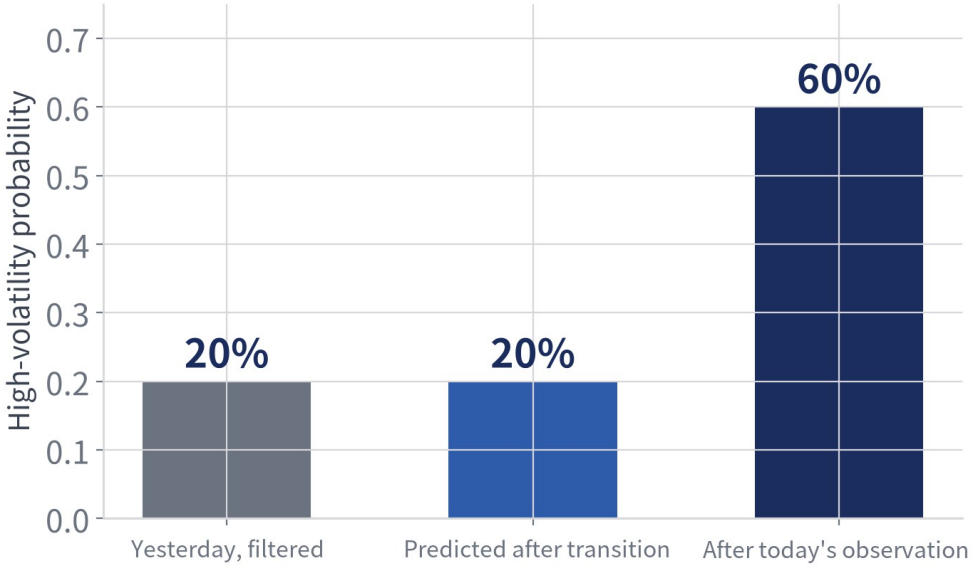
One large move lifts the high-volatility probability to 60%

Yesterday: calm 80%, high volatility 20% · relative emission densities 1 and 6 are assumed to show the structure

$$q_t(C) = 0.8(0.95) + 0.2(0.20) = 0.80$$

$$q_t(V) = 0.8(0.05) + 0.2(0.80) = 0.20$$

$$\boxed{p_t(V) = \frac{0.2(6)}{0.8(1) + 0.2(6)} = 0.60}$$



Choose the HMM algorithm by the question you ask

Same model, different question, different computation

Question	Algorithm	Output
How likely are these data under the model?	Forward	Data likelihood
Knowing data up to today, what is today's regime?	Normalized Forward filtering	Real-time state probabilities
Knowing later data, what was a past regime?	Forward–Backward	Smoothed probabilities
What is the single most likely state path?	Viterbi	Jointly optimal path
What are the transition and emission parameters?	Baum–Welch / EM	Estimated parameters

Forward sums; Viterbi takes the maximum

Unnormalized Forward values — summing the final-time values over states gives the data likelihood

$$\underbrace{\alpha_t(j)}_{\textit{Forward value}} = b_j(r_t) \sum\nolimits_i \alpha_{t-1}(i) a_{ij}$$

Forward

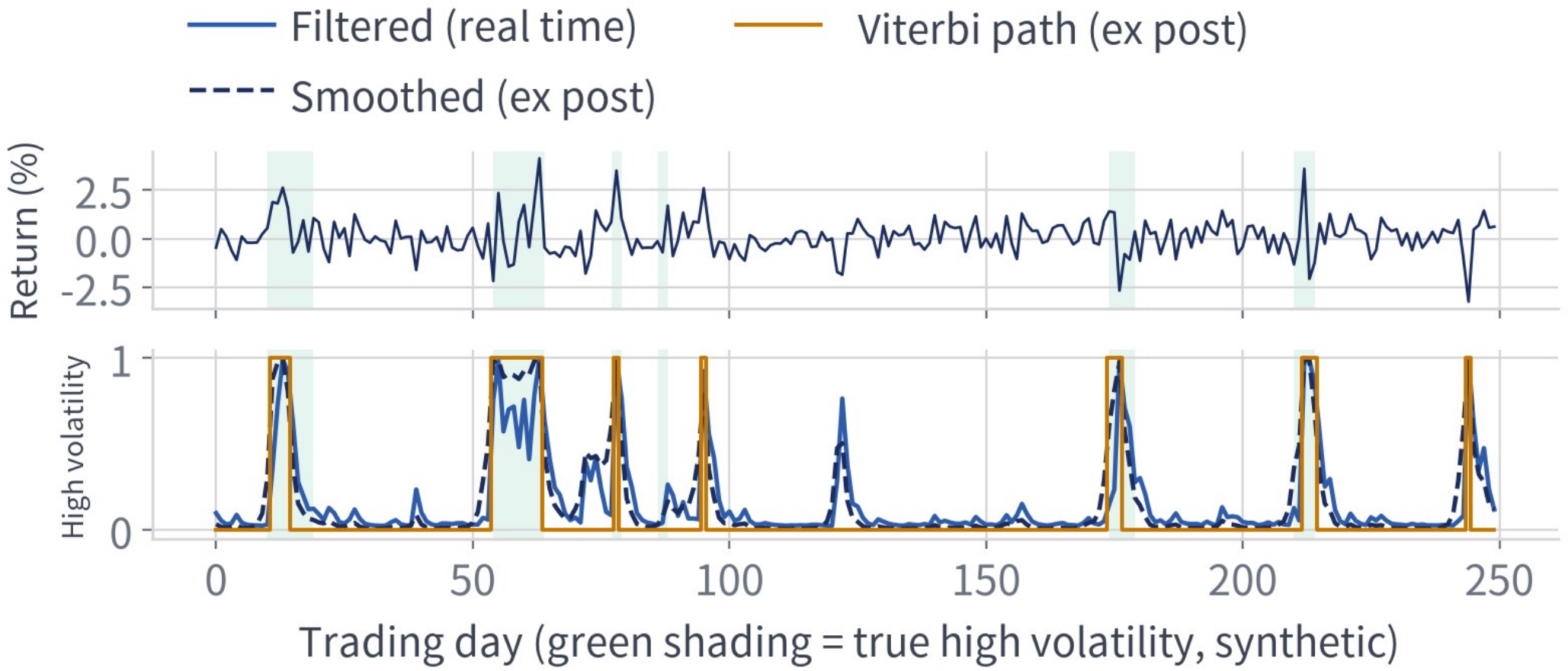
- Long samples require log or scaled computation

Viterbi

- Record the maximum and the back-pointer instead of the sum
- Chaining each day's most probable state ≠ the jointly optimal path

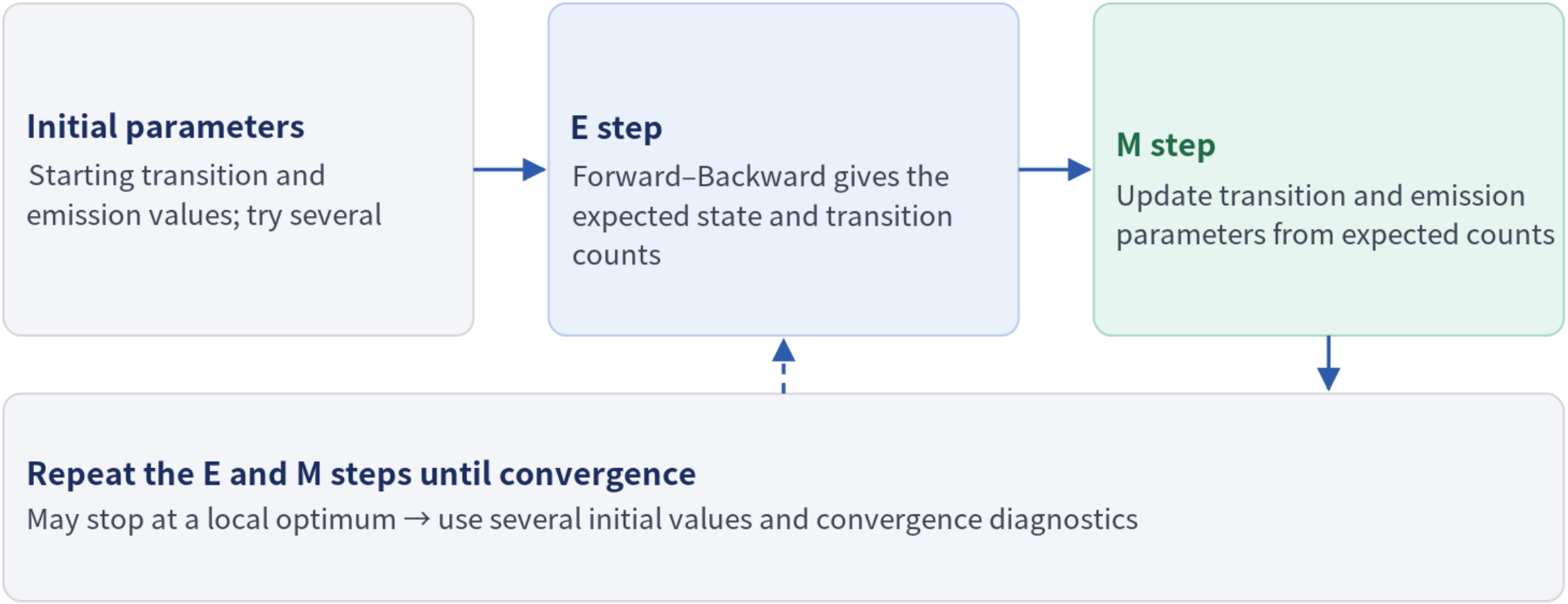
Real-time filtered probabilities lag and are noisier than ex-post probabilities

Synthetic returns from known transition and emission · filtering vs smoothing and Viterbi · ※ synthetic data



Baum–Welch iteratively updates parameters from expected counts

Because states are hidden, first compute expected state and transition counts, then update the parameters



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PART 3 · APPLICATION

Regime Probabilities and Risk Budgets

Rule design · Benchmarks · Limits and extensions

When the high-volatility probability rises, cut the risk budget

Fix the rule in the training period in advance; execute on the next trading day, net of trading costs

Alternatives	Purpose
Constant weight	Simple baseline
Past-volatility threshold	Baseline that needs no HMM
Filtered regime probability	Tests the extra information from combining transition and emission

What 60% means

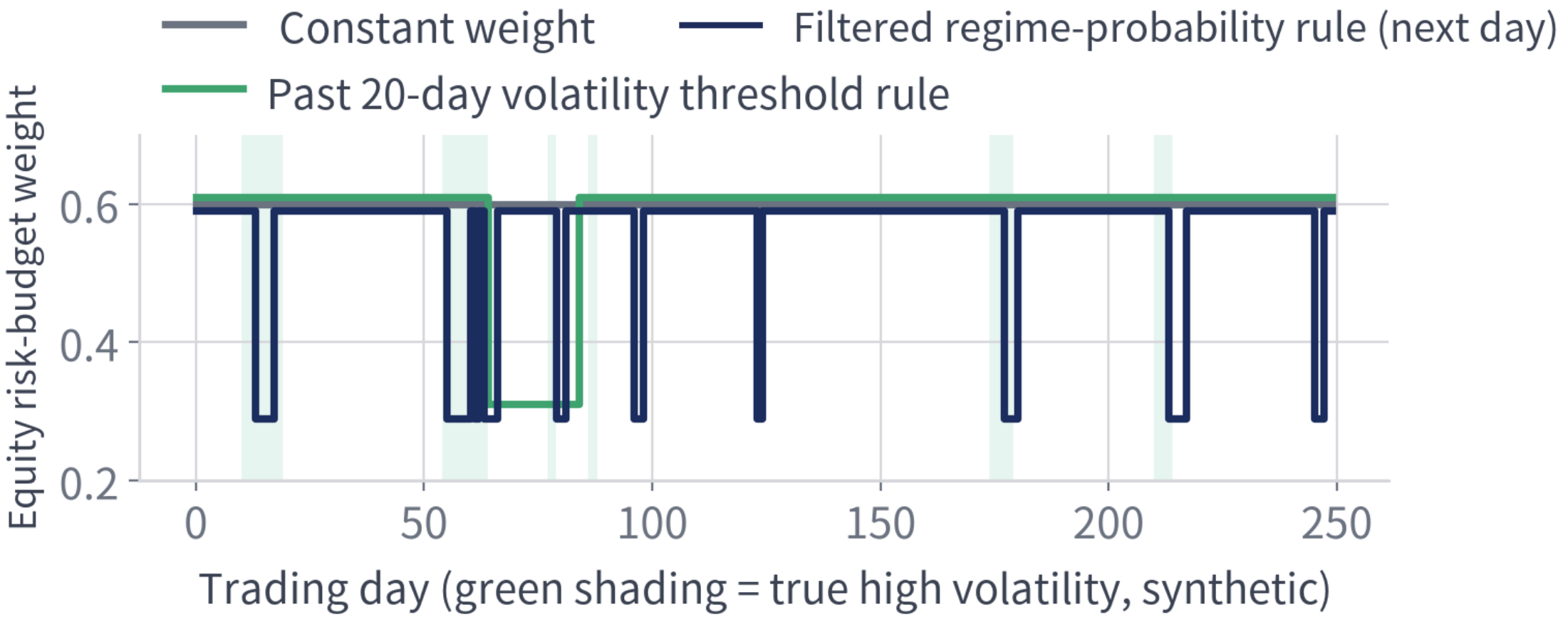
- Neither that today is surely a crisis nor that tomorrow has a 60% chance of falling

No leakage

- Compute today's probability with parameters estimated on past data only

Probability rules change weights often, which can raise costs

Example rule (cut equity weight when the high-volatility probability exceeds 50%) on the same synthetic data · ※ synthetic data



More states do not mean better out-of-sample performance

Limits of the HMM and optional extensions

In-sample fit is not out-of-sample performance — choose the number of regimes and distributions by validation

LIMITS

- A normal mixture alone may not capture extreme tails and serial dependence
- Geometric durations may be inappropriate
- State means have large estimation error and are less stable than variance differences

Optional extensions

- Explicit-duration models
- Regime-specific AR and volatility, Student-t emissions
- Bayesian HMM

The lab goes from a synthetic HMM to a real risk budget

Paste into Claude Code and check the probabilities, transitions and strategy comparison

[Input] Prompt · Claude Code

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Write and run an introductory two-state HMM lab.
First generate synthetic returns from known transition and emission distributions
and save the true states.
Compare Forward filtering, Forward–Backward and Viterbi
in separate figures.
Ensure numerical stability with normalization and log computation.
Train from several initial values and order the state labels
by volatility.
Split the real financial example chronologically into train, validation and test.
Compute real-time filtered probabilities without future observations
or full-sample estimates.
Compare constant weight, a simple volatility rule and the HMM risk-budget rule
under the same costs.
Do not misuse smoothing or Viterbi as real-time signals.
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Check

- State probabilities at each time sum to 1
- Transition matrix rows sum to 1
- True-state accuracy only in the synthetic experiment
- Do not score real data as if the true regime were known

Regimes are categories; judgments are probabilities

Next week: continuous-time shocks and mean reversion

1 Transition + emission

The transition matrix sets regime persistence; the emission distribution sets the shape of returns in each regime.

2 An algorithm per question

Real time: filtering; reinterpreting the past: smoothing; single path: Viterbi; parameters: Baum–Welch.

3 Compare strategy rules under the same costs

Against constant weight and a volatility threshold, assess only the extra information the HMM provides.

Week 10 continuous state (Kalman) ↔ Week 11 discrete state (HMM) — both ‘predict, then correct’